

Malware Detection Pipeline Investigation Report

Prepared By

Mohamed Ibrahim Shire

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Executive Summary

This project simulated a real-world malware delivery and execution attempt against a corporate Windows 10 endpoint. The objective was to validate the organization's ability to detect, investigate, and contain malicious activity using multiple security technologies including Endpoint Central, Sysmon, and Splunk Enterprise.

A malicious executable named **shire.exe** was hosted on a Kali Linux attacker machine and delivered to a Windows endpoint through an internal web server. Upon execution, Endpoint Central immediately detected the threat, blocked execution, and quarantined the file before compromise could occur.

The investigation leveraged Sysmon telemetry, Endpoint Central alerts, and Splunk SIEM correlation to reconstruct the attack chain and validate successful containment.

The incident was classified as **High Severity** due to the attempted execution of a Trojan-class malware sample.

Incident Overview

Incident Type

Malware Execution Attempt

Severity

High

Status

Contained and Remediated

Confidence Level

High

Attack Outcome

Execution Prevented

Malware Classification

Trojan.Generic

Lab Environment

Component	Technology
Attacker Machine	Kali Linux
Victim Endpoint	Windows 10
SIEM Platform	Splunk Enterprise
Endpoint Protection	Endpoint Central
Endpoint Telemetry	Sysmon
Network Segment	192.168.10.0/24

Network Architecture

Kali Linux Attacker

192.168.10.40

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Python HTTP Server

Port 8080

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Windows 10 Endpoint

192.168.10.10

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Endpoint Central Agent

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Sysmon Telemetry

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Splunk Enterprise

Attack Scenario

The objective of the exercise was to simulate malware delivery, execution, detection, and response within a controlled SOC environment.

The attacker hosted a malicious executable named **shire.exe** on a Kali Linux system using a temporary Python HTTP server.

The victim endpoint downloaded the executable and attempted to launch it. Endpoint Central detected the malware immediately and prevented successful execution.

All security events were forwarded to Splunk for centralized investigation.

Attack Timeline

Time	Event
06:35:00	Malware hosted on Kali Linux
06:35:10	Victim downloads shire.exe
06:35:26	User attempts execution
06:35:26	Sysmon Event ID 1 generated

Time	Event
06:35:27	Endpoint Central detects Trojan.Generic
06:35:27	Malware execution blocked
06:35:28	File moved to quarantine
06:35:30	Alert forwarded to Splunk
06:36:00	Investigation initiated
06:40:00	Incident classified and contained

Initial Access Investigation

Malware Hosting Analysis

Evidence confirmed that the malware was hosted using Python's built-in web server functionality.

Command Observed:

```
python3 -m http.server 8080
```

Source Host:

192.168.10.40

Listening Port:

8080

Analysis:

The attacker established a temporary HTTP service to distribute the malicious payload to the victim endpoint.

Malware Delivery Investigation

Indicators Identified

IOC Type	Value
Malware File	shire.exe
Source IP	192.168.10.40
Delivery Port	8080
Threat Classification	Trojan.Generic

Analysis:

The executable was successfully downloaded to the victim endpoint prior to detection.

This stage mirrors common malware delivery techniques frequently observed in phishing and drive-by download campaigns.

Endpoint Detection Analysis

Endpoint Central Alert

The Endpoint Central security engine immediately generated a high-severity alert after execution was attempted.

Detection Details

Threat Name:

Trojan.Generic

Detection Engine:

DeepAV Engine

Severity:

High

Status:

True Positive

Affected Device:

Windows-The-Analyst-Endpoint

Security Control Response

Endpoint Central automatically executed the following actions:

- Identified malicious executable.
- Prevented process execution.
- Generated security alert.
- Quarantined malware.
- Updated incident status.
- Preserved endpoint integrity.

Outcome:

Successful Prevention

No compromise observed.

Sysmon Investigation

Sysmon telemetry provided detailed visibility into endpoint activity.

Event ID 1 – Process Creation

Purpose:

Track process execution activity.

Findings

Process:

shire.exe

Parent Process:

explorer.exe

User:

The-Analyst-PC\LocalAdmin

Execution Time:

2026-05-19 06:35:26

Command Line:

C:\The Analyst\Admin\Downloads\shire.exe

Analysis:

The parent process indicates that the malware was manually executed by the user through Windows Explorer.

This behavior is consistent with user-driven malware execution.

Event ID 11 – File Creation

Purpose:

Track file creation activity.

Findings

The executable was successfully written to disk prior to execution.

Analysis:

This event validates successful payload delivery before security controls intervened.

Event ID 3 – Network Connection

Purpose:

Track outbound network activity.

Findings

Destination IP:

192.168.10.40

Destination Port:

8080

Protocol:

TCP

Analysis:

The malware attempted communication with its originating host immediately after execution.

This behavior is commonly associated with:

- Command and Control communication
- Payload staging
- Remote instructions
- Malware beaconing

Splunk Investigation

Malware Execution Detection

```
index=* EventCode=1 Image="*shire.exe"
```

Purpose:

Identify execution attempts involving the malicious payload.

Outcome:

Execution event successfully identified.

File Creation Detection

```
index=* EventCode=11 TargetFilename="*.exe"
```

Purpose:

Identify executable files written to endpoint systems.

Outcome:

shire.exe creation event located.

Network Connection Investigation

```
index=* EventCode=3 DestinationPort=8080
```

Purpose:

Identify outbound communications related to the malware.

Outcome:

Connection attempt to attacker host detected.

Malware Hunting Query

```
index=* EventCode=1 Image="*shire.exe"  
| stats count by host User CommandLine
```

Purpose:

Determine whether the malware executed on additional systems.

Outcome:

Single endpoint affected.

No lateral spread identified.

Incident Severity Assessment

Category	Value
Incident Type	Malware Execution Attempt
Severity	High
Confidence	High
Impact	Potential Endpoint Compromise
Status	Contained
Risk Level	High

Root Cause Analysis

The incident occurred because the endpoint user was able to download and attempt execution of an untrusted executable.

Contributing Factors:

- User execution privileges.
- Malware delivery over HTTP.
- Absence of application allowlisting.
- User interaction with suspicious files.

Security controls successfully prevented compromise before payload execution completed.

MITRE ATT&CK Mapping

Tactic	Technique	ID
Initial Access	Ingress Tool Transfer	T1105
Execution	User Execution	T1204
Execution	Command and Scripting Interpreter	T1059
Command and Control	Application Layer Protocol	T1071
Defense Evasion	Masquerading	T1036

Containment Actions

Immediate actions performed:

1. Malware execution blocked.
 2. File quarantined automatically.
 3. Endpoint monitored for additional activity.
 4. Incident escalated for review.
 5. IOC information collected.
 6. Threat hunting performed across endpoints.
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Lessons Learned

This project demonstrated the effectiveness of layered security monitoring and automated malware prevention technologies.

Key takeaways:

- Endpoint Central successfully blocked malicious execution.
- Sysmon provided detailed forensic visibility.
- Splunk enabled centralized investigation.
- Threat activity was reconstructed quickly.
- Multiple detection layers reduced response time.

The exercise highlights the value of combining endpoint protection, telemetry collection, and SIEM correlation to defend against malware attacks.

Conclusion

The malware detection pipeline successfully identified, blocked, and contained a Trojan malware execution attempt before compromise occurred.

Evidence from Endpoint Central, Sysmon, and Splunk confirmed the complete attack lifecycle from malware hosting and delivery to execution attempt and remediation.

The investigation demonstrated practical SOC analyst capabilities including malware analysis, endpoint investigation, log correlation, incident response, IOC identification, threat hunting, and security monitoring.

The incident was contained successfully with no evidence of persistence, privilege escalation, lateral movement, or system compromise.